

Exploring Unlabelled Speaker Recognition

Table of Content

1.0 Data Exploration and Analysis.....	3
1.1 Initial steps.....	3
2.0 Proposed Solution(s).....	4
2.1 Data preprocessing.....	4
2.2 Feature Extraction.....	4
2.3 Clustering Algorithms.....	5
2.4 Evaluation.....	6
3.0 Implementation Strategy.....	7
3.1 Data Preprocessing.....	7
3.2 Feature Extraction.....	7
3.3 Clustering Algorithm.....	8
3.4 Evaluation.....	8
4.0 Challenges and Considerations.....	9
4.1 Challenges.....	9
4.2 Mitigations and Considerations.....	10
4.3 Assumptions.....	11

1.0 Data Exploration and Analysis

Performing speaker recognition using unlabelled audio data poses challenges. Working with such audio data relies on the unsupervised learning model, where the speaker's features play a vital role in grouping similar recordings into a cluster. Therefore, the recording quality and characteristics of the over 200 recordings are important to accurately extract the speaker's features for each recording. Some of the speaker's characteristics include pitch, tone and volume.

1.1 Initial steps

Exploration on the features/characteristics of audio data is necessary to understand the patterns of data and identify any potential issues that might negatively affect the training process. The steps include statistical analysis, listening and audio data visualization.

Statistical Analysis

Activities such as identification of missing audio, checking sample rate and measuring duration for each recording are helpful to reveal the data patterns. It helps determine the preprocess steps by flagging broken audio, inconsistent sample rates and unusual long recording. Audio trimming has to be taken into account if the audio duration varies too widely.

Listening to the audio

Randomly pick a subset of the recordings to explore the nature of the data. The speaker's features like tone, pitch, pause and gender can be identified by listening to the recordings, as well as the recording's quality. Some potential challenges that affect the recording quality are background noise, poor recording conditions (microphone hiss), overlapping speech and prolonged pause.

Audio data visualization

Waveforms, MFCCs, and spectrograms are visual representations that capture recording features such as loudness, frequency, and shape. By analysing these visuals, it is possible to spot differences that might distinguish between speakers who have different speaking styles. Recordings from the same speaker would have similar pitch and loudness, whereas the pitch, loudness, and signal shape would differ for recordings from different speakers, even when speaking the same sentences. Moreover, random fluctuations often indicate the presence of noise and distortion while a wide variety of amplitude peaks suggest the necessity of normalization. These visual insights are valuable in making informed decisions on audio preprocessing steps.

Additionally, we can perform clustering using PCA and t-SNE to view the overall cluster and relationships. Both methods reduce dimensionality and measure the distance between recordings, but in different ways. PCA and t-SNE are complementary, providing a comprehensive view on the data structure.

2.0 Proposed Solution(s)

Considering the over 200 unlabelled recordings that belong to 200 speakers, it would be an unsupervised learning problem to cluster similar recordings. The purpose is to automatically recognise which of the 200 speakers is speaking when a new recording is introduced.

The proposed pipeline starts from data preprocessing, feature extraction using pre-trained embeddings, performing clustering and evaluating the performance using internal metrics, visual evaluation and manual checking.

2.1 Data preprocessing

The data and consistency of input recordings directly affect the quality of the extracted speaker embeddings. Variations in sample rate, background noise, silence, and recording conditions may introduce unwanted acoustic variability. This reduces the ability of the embedding model to capture speaker features. Therefore, preprocessing techniques are considered to standardise the recordings before feature extraction.

Based on the data exploration and analysis result, some methods that are considered are noise reduction, voice activation detection (VAD), resampling, and RMS normalization:

1. Noise reduction is applied only when significant noise is detected during exploratory data analysis. Background noise could interfere with feature extraction in embeddings.
2. Voice Activity Detection (VAD) removes silence from recordings so that the embeddings are able to focus only on the human speech.
3. Resampling is necessary if the sample rate of recordings is inconsistent. Standardise the sample rate of all recordings to 16kHz to ensure compatibility with embedding models such as ECAPA-TDNN.
4. RMS normalization may be applied when large differences of amplitude in recordings are observed. The purpose is to reduce the variability caused by volume to allow the embedding model focus on speaker features.

2.2 Feature Extraction

Feature extraction transforms raw audio signals into meaningful representations that capture speaker-specific characteristics. The selection of embeddings is pivotal for unlabelled speaker recognition tasks, as clustering is highly dependent on extracted features. The quality of the extracted embeddings directly influences the ability of clustering algorithms to distinguish between speakers. Two modern pre-trained deep learning-based speech representation models are x-vector and ECAPA-TDNN.

X-vector is a deep neural architecture that produces fixed-length embeddings using Time Delay Neural Networks. It transforms variable-length speech signals into compact fixed-length embeddings that capture speaker-specific characteristics. The statistics pooling layer calculates the mean and standard deviation for each input segment. The x-vector embeddings can be compared by distance or similarity, making them ideal for clustering algorithms.

ECAPA-TDNN is an enhanced TDNN-based speaker embedding architecture that improves upon TDNN and x-vector approaches. It incorporates channel attention mechanisms and multi-scale feature aggregation to enable the network to focus on speaker features. The self-attention mechanism allows the model to emphasize informative speaker-related features, producing embeddings that are more robust under different recording conditions.

Pre-trained speaker embedding models are suitable for unlabelled clustering problems because they have learned meaningful speech representations from large-scale speech datasets, making them a good fit to uncover features of unlabelled recordings. Modern pre-trained models such as x-vector and ECAPA-TDNN were trained with massive data to learn complex feature patterns without requiring labelled data. These embeddings provide a suitable representation for clustering algorithms to identify clusters where similarity between recordings can be measured.

2.3 Clustering Algorithms

After the embeddings have been extracted, clustering algorithms are applied to group similar recordings with similar speaker characteristics. Considering no ground truth labels are available, the clustering algorithms depend entirely on the feature relationships between embeddings. Therefore, multiple clustering algorithms are proposed to experiment and determine which approach produces the best performance – K-Means, Spectral clustering and Agglomerative Hierarchy Clustering (AHC).

K-Means is a simple and efficient clustering algorithm that groups similar recordings into clusters by computing the distance between points. A predefined number of clusters is required, which is 200 corresponding to the number of speakers in this case. K-Means algorithm starts by grouping the embeddings into a predefined number of clusters. Each cluster is represented by a centroid, and new recordings are assigned to the nearest cluster centroid. The iterative process continues until the assignment between iterations stops changing. K-Means can be used as the baseline of partition-based algorithms.

Spectral clustering is another clustering algorithm that groups data points based on similarity and connectivity. The similarity and connectivity between datapoints are calculated using the Graph Laplacian to project the data into a lower-dimensional space. Then, the K-Means algorithm is applied to detect clusters. Hence, spectral clustering is able to identify non-linear and complex shapes or relationships that are suitable for embeddings that have non-linear structures.

Agglomerative Hierarchy Clustering (AHC) is a bottom-up algorithm that starts with every recording as a cluster, and progressively merges the two most similar clusters until a single root cluster is reached. The criterion to select two clusters for merging into one is linkage. Common linkage methods include single, complete, average and Ward linkage determine how similarity is measured during merging. Experimenting and evaluating different linkages may improve the performance to yield more meaningful or compact clusters. Moreover, the cluster hierarchies can be visualised using dendrograms – a tree-structure graph that shows the hierarchy of clusters. The node heights represent the dissimilarity between clusters. AHC can be used with or without having a predefined number of clusters, however, the optimal number of clusters can be determined by looking at the dendrogram.

By experimenting with multiple clustering algorithms, the performance can be evaluated using internal validation metrics. The most well-separated clusters will be selected.

2.4 Evaluation

Evaluating clustering algorithm performance is challenging in the absence of ground truth labels. Since the normal evaluation metrics such as accuracy, precision and recall are not suitable for this case, internal evaluation metrics and visual inspection are used to assess the model performance.

Internal evaluation metrics that can be used are Silhouette Score, Davies-Bouldin Index (DBI) and Calinski-Harabasz Index (CHI). Each metric evaluates a different aspect of performance. Silhouette score measures the similarity of a point to its own cluster to other clusters, ranging from -1 to 1, where a higher score indicates a higher similarity and compactness in a cluster. DBI evaluates the average similarity between clusters, with lower values indicating better clustering performance. CHI, on the other hand, measures the ratio of between-cluster dispersion to within-cluster dispersion. It calculates the distance of data points to their own cluster centroids and global centroid, preferably a higher value indicating dense clusters and well-separated.

Visual inspection is necessary, as internal evaluation metrics are not always aligned with human perception. Visualising the embeddings and clusters – such as dendrogram – provides a complementary way to assess clustering quality. PCA and t-SNE provide high-dimensional embedding visualization to identify outliers and overlapping between clusters.

3.0 Implementation Strategy

Based on the above discussion, the implementation strategy will include data preprocessing, feature extraction, clustering algorithm and evaluation.

3.1 Data Preprocessing

Depending on the data exploration and analysis, some techniques that would benefit speaker clustering are resampling, noise reduction, voice activation detection (VAD) and RMS normalization.

Resampling

The recordings must be at the same sample rate to proceed with embeddings. Hence, consider resampling the sample rate to 16kHz, which is the standard sample rate for most embeddings.

Noise Reduction

Poor audio quality would significantly affect the clustering of recordings, as the embeddings may treat it as one of the features. Perform noise reduction if any background noise is spotted to ensure the recording quality.

Voice Activation Detection (VAD)

Some recordings may contain prolonged pause at the beginning or at the end. Voice Activation Detection (VAD) will be used to remove silence so that human speech is isolated from background noise. This provides a clean recording for models to learn and perform clustering more accurately.

RMS normalization

If the recordings have very different volume levels, normalization is necessary to ensure volume consistency for optimization of clustering algorithms. This prevents extreme amplitude from dominating the training and increases the ability of the model to cluster recordings.

3.2 Feature Extraction

Feature extraction is required to capture human speech before feeding the audio to the model. The preprocessed recordings will be transformed into embeddings. Examples of techniques to extract speech features are ECAPA-TDNN and x-vector.

ECAPA-TDNN

The Speechbrain ECAPA-TDNN embedding will be used to extract speaker embeddings from each recording. Fixed-length embedding vectors will be extracted and used as input for the clustering algorithm.

x-vector

X-vector is a neural architecture for speaker embedding with an embedding dimension of 512. It works with fixed-length audio representation to cluster speech by features. Similarly, speechbrain/spkrec-xvect-voxceleb is used to transform data into embeddings.

Both ECAPA-TDNN and x-vector are implemented and evaluated to determine which model produces better embedding representations for clustering. The embeddings will be visualised using PCA and t-SNE. Clear boundaries and less overlap between clusters would indicate a better embedding quality. Therefore, increased overlap compared to the data exploration and analysis stage may suggest poor embeddings or presence of noise, indicating that changing the embedding model or revisiting preprocessing steps may be necessary.

3.3 Clustering Algorithm

After converting the audio signals into embeddings, unsupervised learning will be implemented to cluster recordings into groups based on their features.

K-Means

K-Means, a simple and fast model that can cluster recordings into a number of groups, which is 200 in this case, will be used as a baseline clustering method. The `n_clusters` parameter will be set to 200. The embeddings from the feature extraction will be used as input for the K-Means algorithm.

Agglomerative Hierarchical Clustering (AHC)

AHC algorithm group recordings by merging the most similar pairs iteratively. Similarly, the `n_clusters` is set to 200. Then, feed the embeddings to the algorithm to perform clustering based on the features extracted. Different linkage methods and metrics will be experimented to determine the best configuration. Then, dendrograms are generated to visualise the hierarchy of clusters.

Spectral Clustering

While K-Means and AHC compute the distance between points, spectral clustering finds patterns based on connectivity. It measures the relationships between clusters to identify the cluster with stronger interconnectivity. The number of clusters is set to 200 representing 200 speakers. Different affinity values will be tested to compare the effect on clustering performance.

3.4 Evaluation

Several metrics will be used to evaluate clustering performance without ground truth speaker labels in order to choose the best fit model.

Internal Evaluation Metrics

Silhouette score, Davies-Bouldin Index (DBI) and Calinski-Harabasz Index (CHI) will be evaluated for each clustering algorithm. Better clustering performance is indicated by a higher Silhouette score and CHI value, with a lower DBI score. This suggests better cluster separation and compactness.

Visual Evaluation

Another technique to manually evaluate the clustering performance is by observing the PCA and t-SNE visualizations of the cluster structure. Both PCA and t-SNE provide visualization of clustering performance. An ideal result would be less overlapping between clusters.

Manual Verification by Listening

Select some clusters and listen to a subset of randomly chosen recordings within a cluster. Identify any obvious recordings from different speakers that have been clustered into groups they do not belong to.

Finally, evaluate different combinations of embeddings and clustering models to figure out the best-performing model to work with over 200 unlabelled recordings using the evaluation method discussed above. The best-performing combination would be selected for the final speaker recognition task.

4.0 Challenges and Considerations

Dealing with more than 200 unlabelled recordings without ground truth speakers introduces challenges at every stage – from data preprocessing and feature extraction to clustering. These potential issues need to be carefully considered when implementing the proposed solutions.

4.1 Challenges

Some challenges that might be inherent in performing speaker recognition are recording quality, identical speaker, variability of speaker and clustering algorithm.

Recording Quality

One of the primary challenges is the recording quality across the 200 recordings. Although the recordings are microphone recorded, the recordings may suffer from inconsistent background noise, microphone hiss, and different sample rate and volume. Moreover, if a recording contains more than one speaker, the features from the second speaker might distort the feature extraction of the targeted speaker. Poor recording quality might contribute to weak features extraction, misleading embedding to treat noise as part of the feature. Consequently, the clustering algorithm may fail to group recordings based on the patterns.

Identical Voices

Small differences between similar voices may be difficult for embedding models to distinguish. The challenge lies in separating speakers whose voice characteristics are highly

similar in terms of pitch, tone and speaking style. Even pre-trained embeddings may not have learned the subtle differences needed to distinguish between such voices, resulting in misclustering of recordings.

Variability of Speaker Characteristics

A speaker's voice can change over time due to illness, aging or emotional state. If the recordings are collected across a long period, the voice features of a speaker may vary due to these circumstances, causing the embeddings or clustering model to miscluster recordings. Additionally, even if all the recordings are captured on the same day, there may still be slight differences in pitch, tone and speaking rhythm. When the recordings of a speaker vary too much, the embedding may fail to cluster them together.

Clustering Algorithm

Different clustering algorithms have inherent assumptions and limitations that may not align with the characteristics of speaker embeddings. Choice of clustering algorithms have a significant impact.

K-Means clustering assumes the number of clusters is defined, which is 200 in this case and it is sensitive to outliers and noise. While over 200 recordings are presented, K-Means may face challenges dealing with large datasets that the computational cost could be expensive.

Spectral clustering requires a high computational cost to calculate the Laplacian. The larger the dataset, the higher the computational cost. Moreover, it is highly dependent on the hyperparameter choices. Proper parameter selection is required to ensure a reliable result.

Agglomerative Hierarchy Clustering (AHC) works well for small and medium-sized datasets, but may struggle with large datasets. Same as K-Means, AHC is sensitive to outliers and noise, thus there is a possibility of overclustering.

Difficulty in Evaluation

The challenge becomes greater when there is no ground truth speaker label to identify any misclustering. Recordings belonging to the same speaker may be failed to group together due to above matters – recording quality, identical voices, variability of speaker characteristics and clustering algorithm – it is difficult to evaluate the clustering accuracy.

4.2 Mitigations and Considerations

While the challenges outlined above are significant and must be taken into consideration, several mitigations can be implemented to improve the likelihood of successfully distinguishing speakers. These include preprocessing techniques, embedding and clustering model selection, and leveraging datapoints to detect the speakers.

Preprocessing Techniques

To address recording quality issues, consider applying preprocessing steps such as noise reduction, voice activation detection (VAD), RMS normalization and resampling to a consistent sample rate. These activities ensure the recording quality and help embeddings capture speaker characteristics rather than noise. For long duration recordings, segment

them into smaller frames to ease feature extraction, and apply windowing technique to ensure the smoothness of audio.

Embedding Selection

Deep learning embedding models, including ECAPA-TDNN and x-vector, leverage neural networks to capture speaker's characteristics. These embedding models have the ability to capture more detailed characteristics of a speaker, improving the separability between similar voices. In addition, we can visualise the clusters of embedding using PCA and t-SNE to identify any potential overlap between clusters.

Model Selection

Selecting a suitable clustering model is vital to cluster recordings based on features. The best approach would be to experiment with different models, starting with K-Means as a baseline, then evaluating performance with AHC and Spectral clustering. Unsupervised clustering algorithms are evaluated via internal validation metrics – Silhouette Score, DBI and CHI. These metrics provide insights into the clustering performance and could suggest further improved robustness. Furthermore, visual inspection via PCA and t-SNE is encouraged to inspect the clusters.

Evaluation

Use internal evaluation metrics, visual evaluation and manual verification to assess the clustering performance due to the absence of ground truth labels. These techniques provide additional information on assessing the embedding and clustering combination.

4.3 Assumptions

Several assumptions are made about the dataset. First, we assume that all the recordings are not severely corrupted. Secondly, the recordings are generally consistent in sample rate, volume level and in duration that the recordings are not too long or too short. While some variability is expected, it should remain within manageable limits for preprocessing to correct. Third, at least 2 recordings are available for each speaker so that the embedding is able to capture meaningful features of all 200 speakers. Finally, we assume that only one speaker's voice is captured in every single recording.